

2022

DJI RC Plus
Repair Training

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THE FUTURE OF POSSIBLE

01 Product Introduction

01 Product Introduction

Product Name: DJI RC Plus for T40/T20P

Product Series: Agras Product

Supported Aircraft: T40/T20P

Image Transmission: O3 Pro

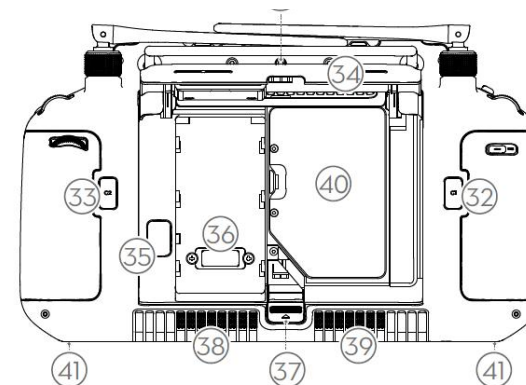
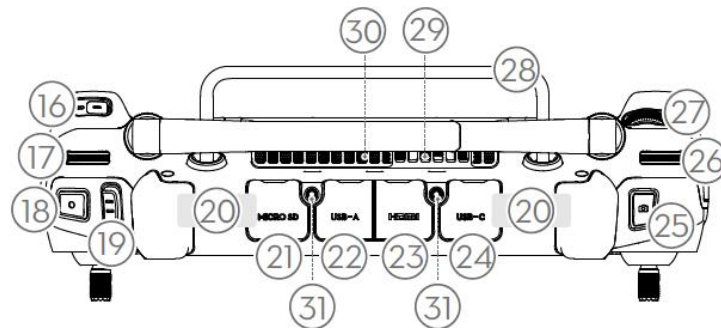
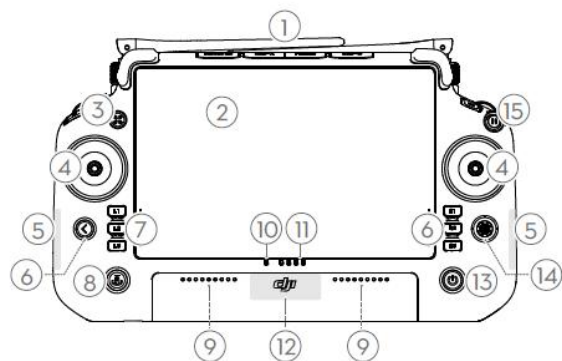
Transmission Range: 5 km (SRRC); 4 km (MIC/KCC/CE); 7 km (FCC)

Frequency Band: 5.8 GHz and 2.4 GHz



High-Brightness Touchscreen	Multiple Wireless Connection	Audio and Video Processing	Extended Storage Capability	More Features
7.02-inch 1920×1200p High-Brightness Touchscreen - Maximum Brightness: 1200 cd/m² - Android System	- Wi-Fi - Bluetooth 4G Dongle	- Built-in Speaker - H.264 4K/60fps and H.265 4K/60fps Video Formats - HDMI 1.4 Port	64 GB Internal Storage - microSD Card	- Reliable in More Environments: -20°C-50°C - Fast-Charging and Long Operating Time - IP54

01 Product Introduction



1. External RC Antennas
2. Touchscreen
3. Indicator Button ^[1]
4. Control Sticks
5. Internal Wi-Fi Antennas
6. Back/Function Button
7. L1/L2/L3/R1/R2/R3 Buttons
8. Return to Home (RTH) Button
9. Microphone
10. Status LED
11. Battery Level LEDs
12. Internal GNSS Antennas
13. Power Button
14. 5D Button

15. Flight Pause Button
16. Customizable C3 Button
17. Left Dial
18. Record Button ^[1]
19. Flight Mode Switch
20. Internal RC Antennas
21. microSD Card Slot
22. USB-A Port
23. HDMI Port
24. USB-C Port
25. Shutter Button ^[1]
26. Right Dial
27. Scroll Wheel
28. Handle

29. Speaker
30. Air Vent
31. Reserved Mounting Holes
32. Customizable C1 Button
33. Customizable C2 Button
34. Rear Cover
35. Battery Release Button
36. Battery Compartment
37. Rear Cover Release Button
38. Alarm
39. Air Intake
40. Dongle Compartment
41. 1/4" Threaded Holes

01 Product Introduction—T30 Remote Controller VS DJI RC Plus



T30 Remote Controller



DJI RC Plus

Materials Available for both DJI RC Plus and T30 Remote Controller:

Material Number	Material Name
YC.WJ.L00051	Cross Self-tapping Screw ϕ 2.0x4.5 (CUK)
YC.WJ.L00178	M4 Locknut
YC.WJ.TT000152.03	Dial Extension Spring
YC.WJ.ZZ002094.02	Dial Decorative Piece

Accessories Available for both DJI RC Plus and T30 Remote Controller:

Material Number	Material Name
CP.BX.000229	CrystalSky WB37 Intelligent Battery
CP.RN.00000230.02	CrystalSky WCH2 Charging Hub Single Unit (RH)
BC.AG.SS000229.03	RTK Dongle

01 Product Introduction—Product Comparison

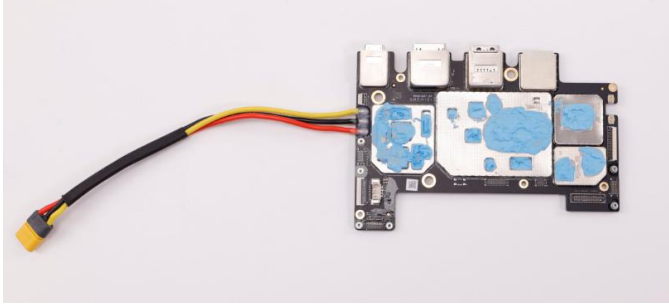
Remote Controller	T30 Remote Controller	DJI RC Plus
Weight	Approx. 630 g	Approx. 1.37 kg (with external battery), 1.2 kg (without external battery)
Working Time	2.5 hours	3.3 hours with internal battery and up to 6 hours with external WB37 battery used
Charging Time	2 hours	2 hours (using a USB-C power adapter with maximum rated power of 65W and maximum voltage of 20 V)
Max Transmission Distance	7 km (FCC), 4 km (MIC/KCCCE), 5 km (SRRC)	5 km (SRRC), 4 km (MIC/KCC/CE), 7 km (FCC)
Operation Temperature Range	0°C-40°C	-20°C-50°C
Screen Brightness	1000 cd/m ²	1200 cd/m ²
Screen Resolution/Frame Rate	1920x1080p, 60fps	1920x1200p, 60fps
Dimensions	5.5-inch	7.02-inch
Storage (ROM)	32 GB	64 GB



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02 Material Identification

02 Material Identification



Core Board

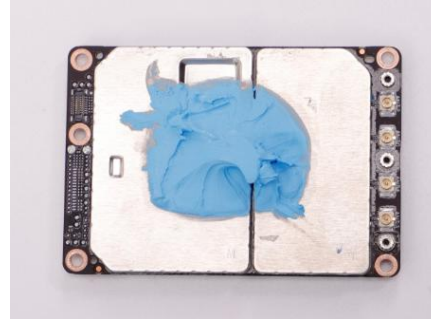
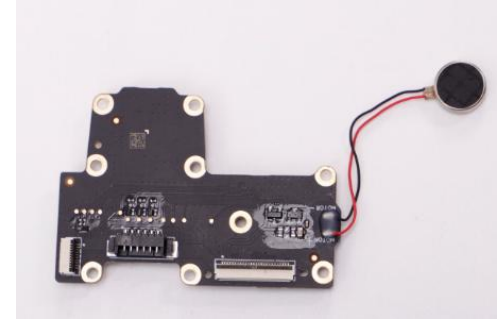


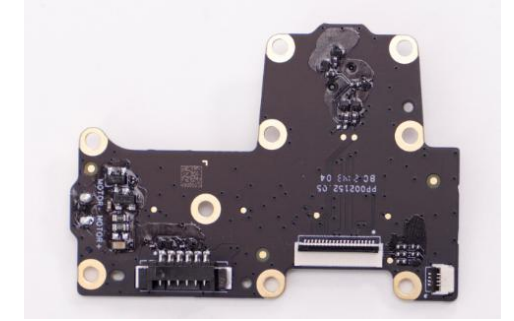
Image Transmission Board

Receive the aircraft signal and transmit the remote controller control signal.

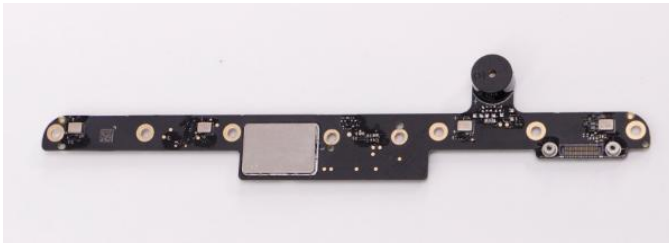


Button Board (Left)

Include the vibration motor.



Button Board (Right)



GPS Board

Include GPS module and buzzer.



Shutter/Pause Button Flat Cable



Flight Mode
Switch/Indicator
Button/Record Button Flat
Cable



4G Dongle Flat Cable

02 Material Identification



Internal Wi-Fi Antenna (Right)



Internal Wi-Fi Antenna (Left)



+



External RC Antenna



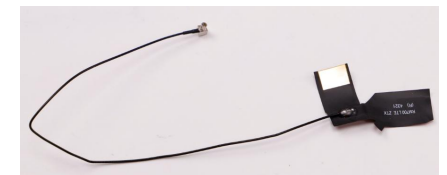
Internal SDR Antenna (Left)



Internal SDR Antenna (Right)

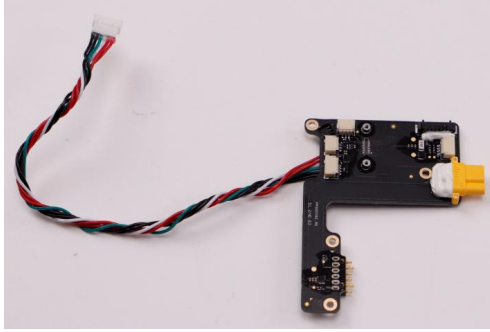


Reflection Board

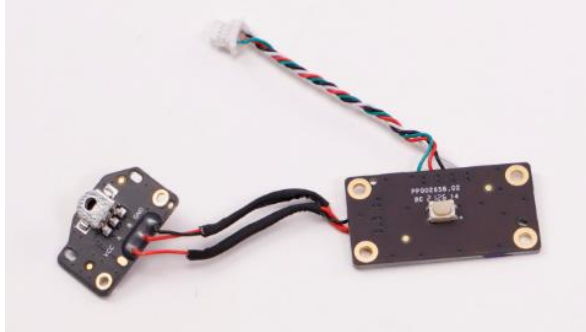


Internal RC Antenna (Right)
(Sticker)

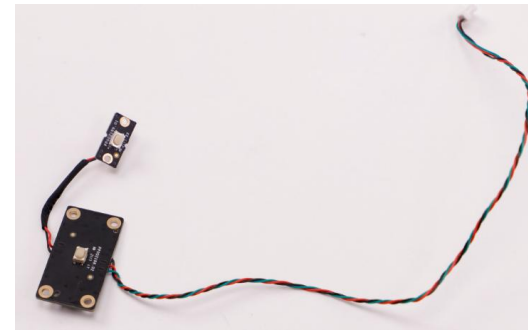
02 Material Identification



Power Port Board



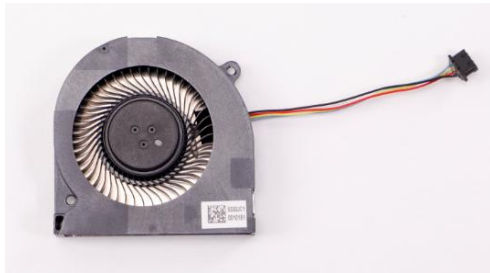
C1-C3 Button Board



C2-Camera Settings Button Board



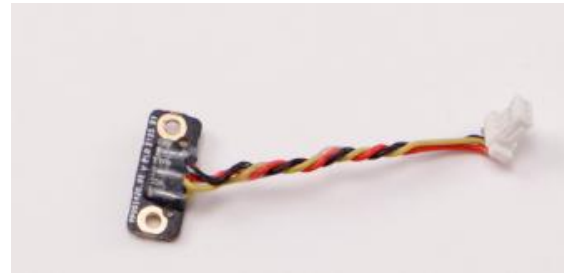
Speaker



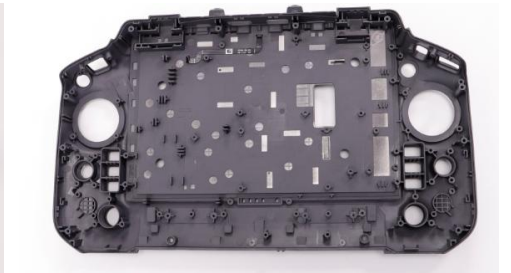
Fan



Left Dial Board



Right Dial Board



Front Shell Module



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03 Repair Level

03 Repair Level

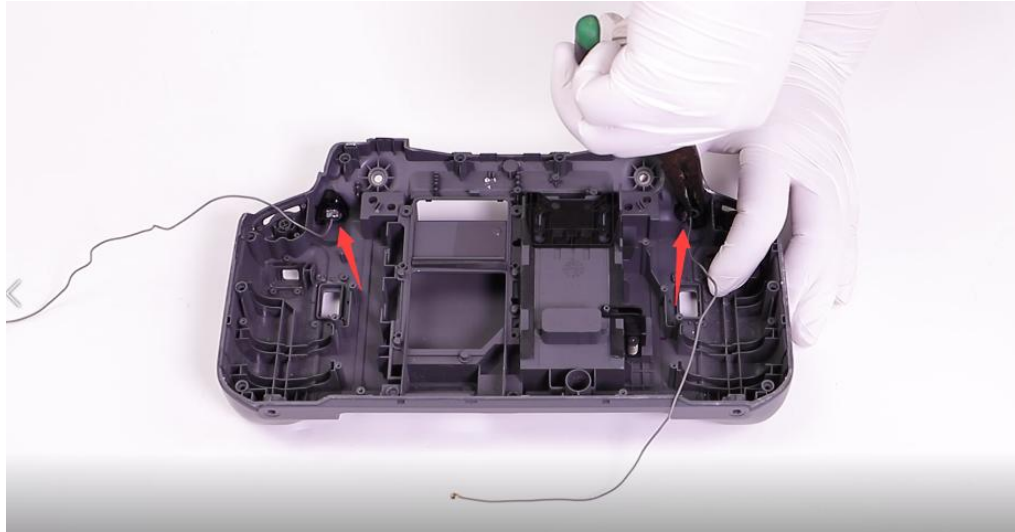
Product	Material	Dealer	Remarks
DJI RC Plus	Remote Controller	Repair it (If the screen is damaged, replace the BC.AG.SS000595.01 front shell module (with the screen).	The air tightness test is not required.



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04 Disassembly and Assembly Notes

04 Disassembly and Assembly Notes—Removing External Antenna and Antenna Base

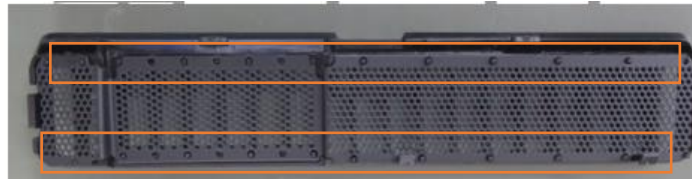
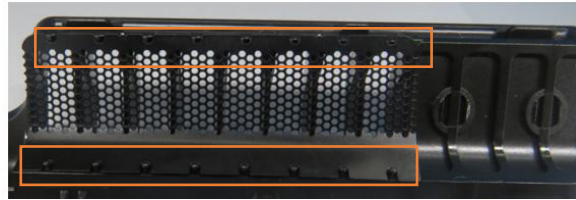


Use the needle nose pliers to hold the buckle of the external antenna, and push it out to remove the external antenna.



The external antenna base is attached by using instant adhesive. The antenna base can be removed by heating with a heat gun after unscrewing the fixing screw.

04 Disassembly and Assembly Notes—Mounting Air Intake/Vent Dustproof Mesh



1. Take the air intake/vent cover and dustproof mesh.

2. Attach the dustproof mesh to the air intake/vent cover, and iron out the locating rods as shown in the orange boxes with the soldering iron so that the dustproof mesh is attached firmly to the cover.

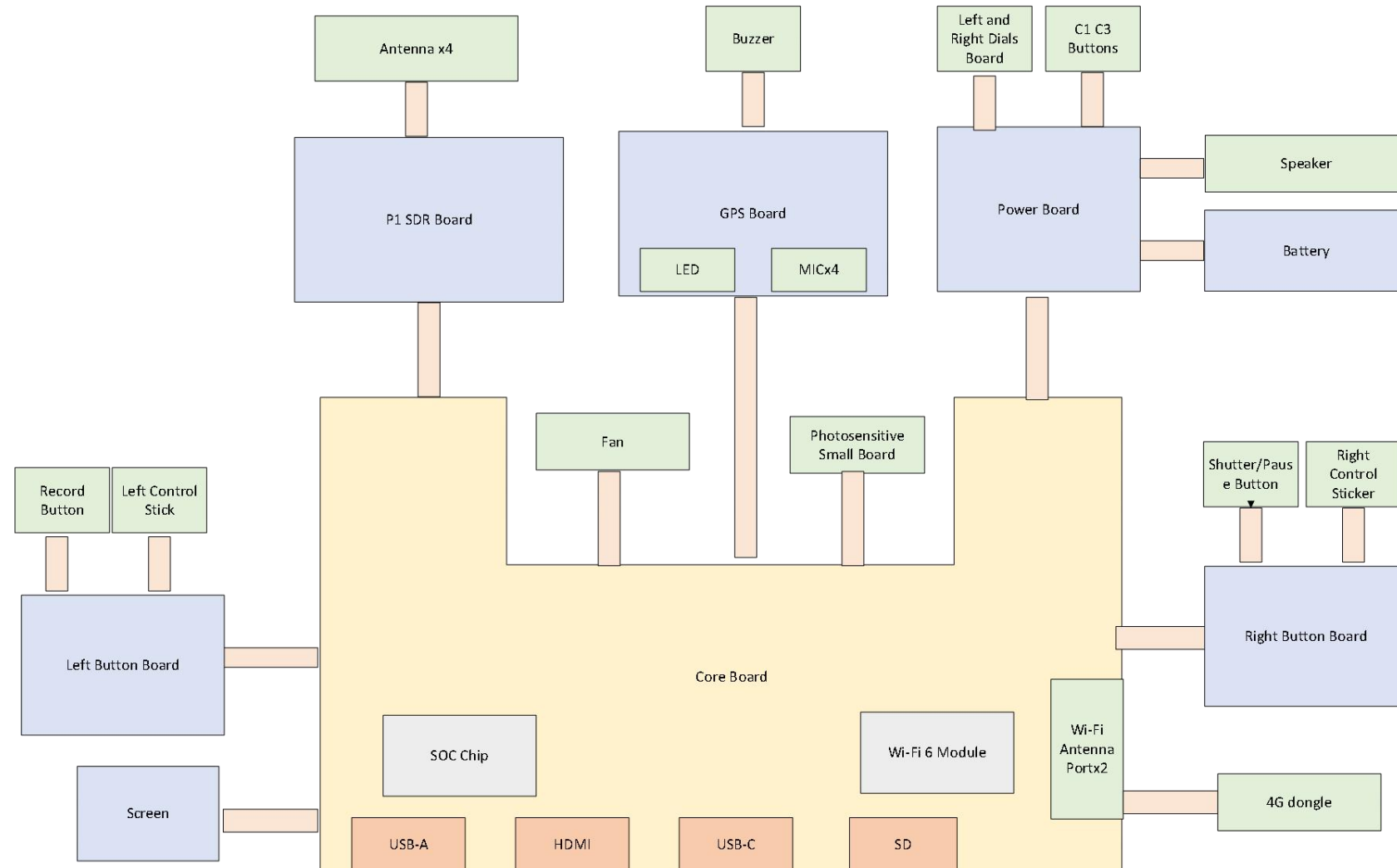
3. The dustproof mesh is attached to the cover as shown.



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05 Principles Diagram and Boards Introduction

05 Principles Diagram





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06 Test Plan and Fixture

06 Test Plan and Fixture

Module	Test Item	Content	Device/Fixture	Test Environment	Test Software (Name/Version)	Damage Assessment Required	Repair Required
Remote Controller	Firmware Update	Update the remote controller firmware to the latest version.	/	/	Official DJI Assistant 2	No	Yes
	App Functionality Test	Check the function by using the app.	/	/	Built-in App	Yes	Yes

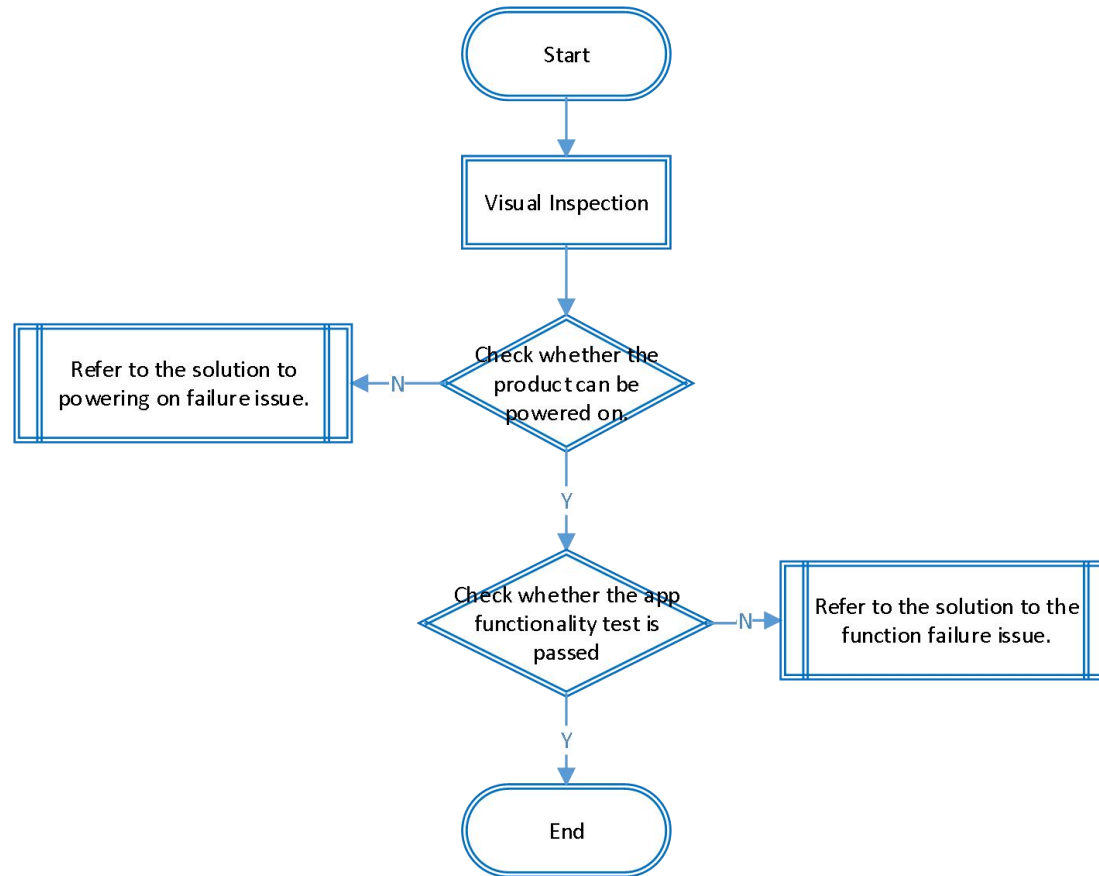


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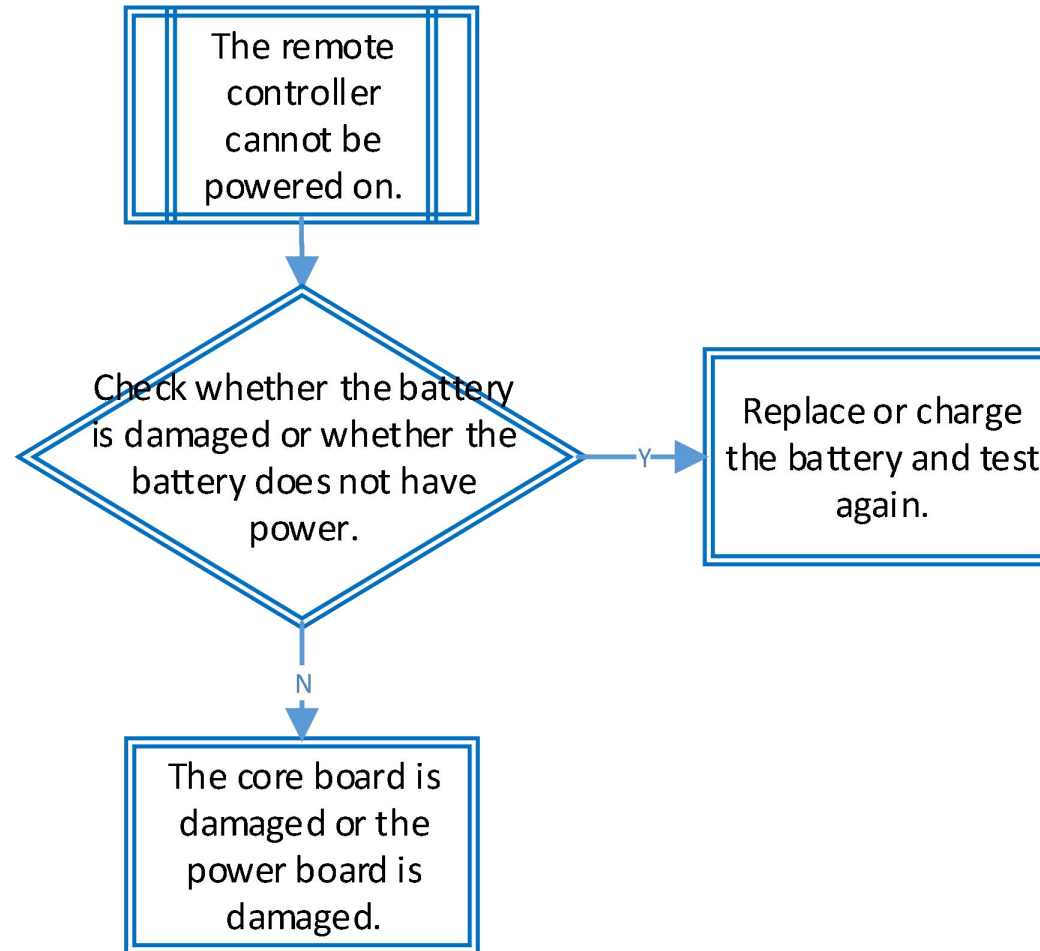
07 Malfunction Assessment and Analysis

07 Malfunction Assessment—RC Malfunction Troubleshooting Main Flow Chart

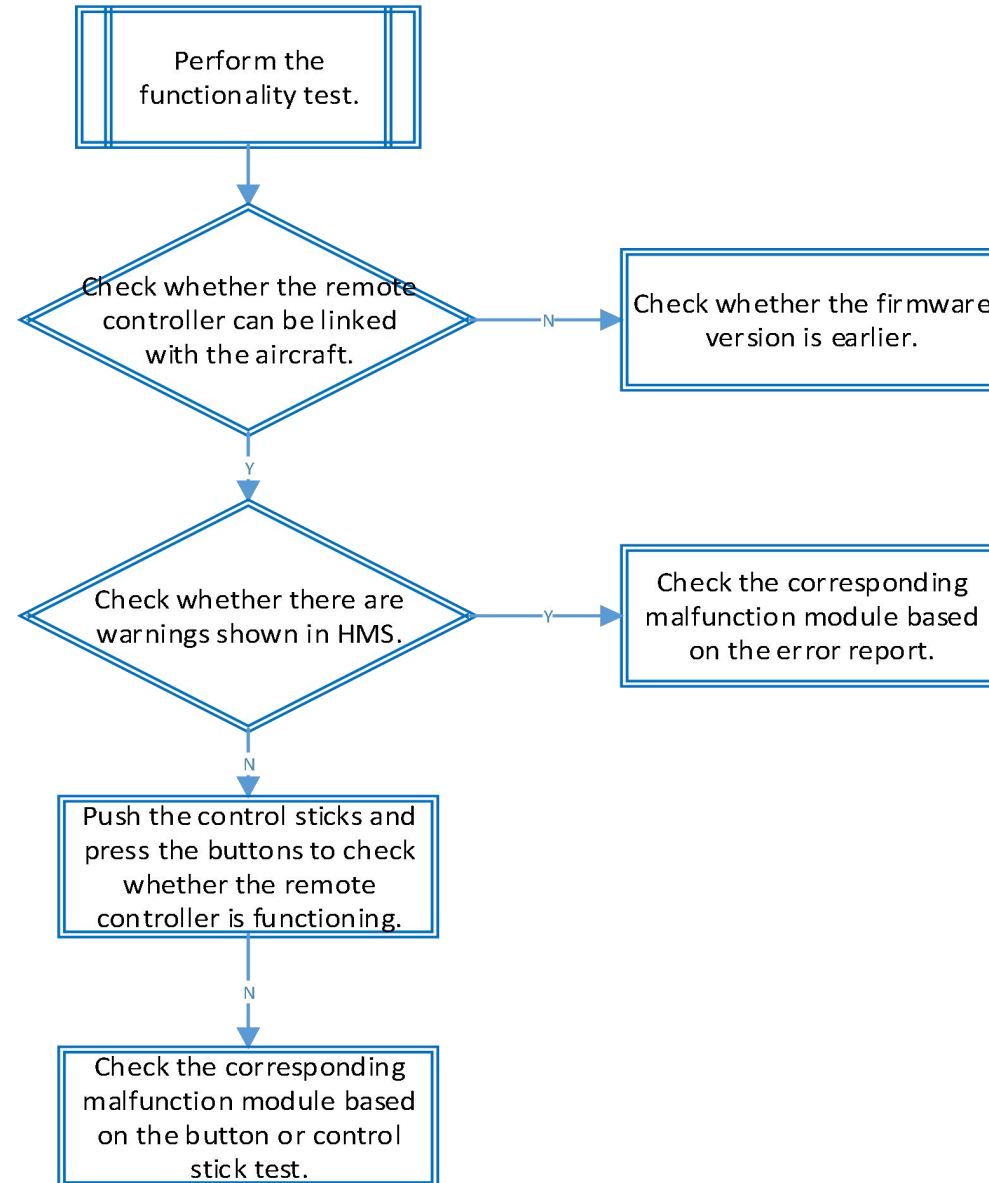
Use the app to perform damage assessment.



07 Malfunction Assessment—Powering on Failure



07 Malfunction Assessment — Function Failure



07 Malfunction Assessment — App Error Report

No.	App Warnings	Solution
1	Firmware update required/Firmware versions do not match. Update firmware.	Use the official DJI Assistant 2 to update the firmware to the latest version.
2	Remote controller battery level low	Charge the remote controller.
3	Remote controller battery initialization error	There is an error with the remote controller battery, so replace the battery and test again.
4	Control sticks not centered	Use the app or perform the link test to calibrate the remote controller.



Thank You!